

REMIND Lab - EmergencyReport Rules

Version 0.1 | Lifecycle, invariants and open decisions

1. Required Creation Data

An EmergencyReport must be created with enough information to characterize and route the occurrence.

Field	Creation rule
title	Required.
description	Required.
location	Required. MVP location is descriptive; automatic geolocation is out of scope.
category	Required. One of the supported EmergencyCategory values.
severity	Required.
status	Not supplied by the user. Defaults to OPEN.
reportedBy	Derived from the authenticated User.
createdAt	Generated by the system.
assignedCompany	Initially empty.
resolvedAt / cancelledAt	Initially empty.

2. Lifecycle

Operation	Current state	Result	Authority
create	-	OPEN	Authenticated User
accept / assume	OPEN	IN_PROGRESS	Eligible Company
cancel	OPEN	CANCELLED	Author User
cancel	IN_PROGRESS	CANCELLED	Author User
resolve	IN_PROGRESS	RESOLVED	Assigned Company
change severity	OPEN	Same status	Author User
change severity	IN_PROGRESS	Same status	Author User

3. Invariants

- Every newly created EmergencyReport starts as OPEN.
- Required creation data cannot be absent.
- The user does not choose the initial status.
- After creation, title, description, location, and category are treated as immutable in the MVP.
- Severity remains editable by the author while the report is active (OPEN or IN_PROGRESS), because real-world urgency may evolve.

- Only OPEN reports may be assumed by a company.
- Only a company eligible for the report category may attempt to assume it.
- Only one company may be responsible for an occurrence at a time.
- Only the assigned company may resolve an IN_PROGRESS report.
- The author may cancel the occurrence in OPEN or IN_PROGRESS.
- RESOLVED and CANCELLED are terminal states.
- Closed reports cannot be reopened. A new occurrence requires a new EmergencyReport.
- Cancellation preserves the occurrence and its history.
- Deletion and cancellation are different concepts. DELETE must not be treated as a synonym for CANCEL.

4. Status and Severity Are Independent

Status answers: where is this occurrence in the response lifecycle? **Severity** answers: how urgent is the situation currently? Changing severity does not itself move the report between OPEN, IN_PROGRESS, RESOLVED, or CANCELLED.

5. Known Engineering Pressure: Concurrent Acceptance

Multiple eligible companies may see the same OPEN report and attempt to assume it at nearly the same time. The business invariant is already clear: only one company can succeed. The technical concurrency strategy is deliberately deferred until architecture and persistence are selected.

6. Open Decisions

- Whether severity may decrease as well as increase.
- Exact Severity values and semantics.
- Exact structure and required fields of ResponseReport.
- Whether and under what circumstances administrative deletion is allowed.
- Whether cancellation of an IN_PROGRESS occurrence requires a reason.
- Whether companies continue to see cancelled reports in operational history and what metadata is retained.